

A Period-Two Counterexample to the Spectral-Envelope Question in arXiv:1012.4549

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Status. Candidate full counterexample to the literal question, likely valid, pending human review. The example uses a nonconstant periodic minimal sequence; an aperiodic or Thue–Morse-only variant is not settled.

Source Question

Lawton’s paper [1] defines, for a minimal characteristic sequence χ_F , its symbolic orbit closure $(X(F), \sigma)$, an invariant ergodic probability ζ , and

$$X_1(F) = \{b \in X(F) : b(0) = 1\}.$$

For $g \in L^2(X(F), \zeta)$ supported in $X_1(F)$, Herglotz’s theorem gives a positive measure ν_g with

$$\nu_g(e_j) = (\sigma^j g, g), \quad j \in \mathbb{Z},$$

and the paper sets

$$\Sigma(F, \zeta) = \{\nu_g : g \in L^2(X(F), \zeta), \text{ supp}(g) \subseteq X_1(F)\}.$$

Question 3.4 on source PDF page 5 asks:

Is $S_e(F) \subseteq \Sigma(F, \zeta)$?

Here $S_e(F)$ is the set of extreme points of the spectral envelope

$$S(F) = \overline{\{|f|^2 : f \in P(F)\}}^{w^*},$$

where $P(F)$ is the set of norm-one trigonometric polynomials whose frequencies lie in F .

this is the case then for $\mu(B) > \frac{1}{2}$, $\alpha(B, F) > 0$ and $\{F, 1 + F, 2 + F\}$ is a Riesz cover for B .

Suggestions for Further Research Theorem 3.2 shows that characterization of the set of extreme points $S_e(F)$ in the spectral envelopes of integer subsets F such that χ_F is a minimal sequence is crucial to understanding the FCE. For such a set F consider the dynamical system $(X(F), \sigma)$ where $X(F)$ is the orbit closure of χ_B . Then $X(F)$ has at least one shift invariant ergodic probability measure ζ .

Remark 3.3. If χ_F is the Thue–Morse minimal sequence then ζ is unique [8].

Define $X_1(F) = \{b \in X(F) : b(0) = 1\}$. For $g \in L^2(X, \zeta, \sigma)$ define $\sigma g(x) = g(\sigma x)$. Then the sequence $(\sigma^j g, g)$ is positive definite so by the Herglotz theorem there exists a positive measure $\nu_g \in M(\mathbb{T})$ such that $\nu_g(e_j) = (\sigma^j g, g)$, $j \in \mathbb{Z}$. Define the set

$$\Sigma(F, \zeta) = \{\nu_g : g \in L^2(X, \zeta), \text{ support}(g) \subseteq X_1(F)\}.$$

The Birkhoff ergodic theorem can be used to show that $\Sigma(F, \zeta) \subset S(F)$. This fact, together with the fact that $Q(F)$ contains no extreme points, suggests research to answer the

Question 3.4. Is $S_e(F) \subseteq \Sigma(F, \zeta)$?

The fact that generalized characters play a crucial role in characterizing the structure of the Banach algebra $M(\mathbb{T})$ suggests research to investigate their utility for characterizing spectral envelopes.

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Proof Intuition

A period-two frequency set separates the two constructions in the question. On the symbolic side, support inside $X_1(F)$ selects only one point of the two-cycle. Its correlations therefore have a fixed phase: they produce only the unshifted uniform measure on the two square roots of 1.

On the spectral-envelope side, phases of polynomial coefficients can move a Fejér peak continuously. Pulling such a peak back through the doubling map produces the uniform measure on any translated two-point orbit. Every such orbit measure is extreme. Choosing the orbit $\{1/4, 3/4\}$ therefore gives an extreme spectral measure that the symbolic correlation class misses.

Counterexample

Theorem 1. *Question 3.4 in [1] has a negative answer as written. There is a nonconstant minimal characteristic sequence χ_F and an invariant ergodic probability ζ on $X(F)$ such that*

$$S_e(F) \not\subseteq \Sigma(F, \zeta).$$

One may take $F = 2\mathbb{Z}$.

Proof. Let $F = 2\mathbb{Z}$. Write $x_0 = \chi_{2\mathbb{Z}}$ and $x_1 = \sigma x_0$. The orbit closure is the two-cycle

$$X(F) = \{x_0, x_1\}, \quad \sigma x_0 = x_1, \quad \sigma x_1 = x_0.$$

It is minimal, and its unique invariant probability is

$$\zeta = \frac{1}{2}(\delta_{x_0} + \delta_{x_1}).$$

Since $x_0(0) = 1$ and $x_1(0) = 0$, we have $X_1(F) = \{x_0\}$.

We first calculate $\Sigma(F, \zeta)$. If g is supported on $X_1(F)$, write $g(x_0) = c$ and $g(x_1) = 0$. Directly on the two-cycle,

$$(\sigma^j g, g) = \begin{cases} |c|^2/2, & j \text{ even}, \\ 0, & j \text{ odd}. \end{cases} \quad (1)$$

Let

$$m_0 = \frac{1}{2}(\delta_0 + \delta_{1/2}).$$

Its Fourier moments satisfy $m_0(e_j) = 1$ for even j and 0 for odd j . By uniqueness of Fourier coefficients of finite measures, (1) gives

$$\Sigma(F, \zeta) = \{am_0 : a \geq 0\}. \quad (2)$$

Now define

$$m_* = \frac{1}{2}(\delta_{1/4} + \delta_{3/4}).$$

We show first that $m_* \in S(F)$ without using any dilation convention. For $N \geq 1$, put

$$f_N(t) = \frac{1}{\sqrt{N}} \sum_{k=0}^{N-1} e^{2\pi i k(2t-1/2)}, \quad t \in \mathbb{T}.$$

All frequencies of f_N belong to $2\mathbb{Z}$, and orthogonality gives $\|f_N\|_2 = 1$. Its squared modulus is the Fejér kernel pulled back by $t \mapsto 2t - 1/2$. Hence for every $\varphi \in C(\mathbb{T})$,

$$\int_{\mathbb{T}} \varphi(t) |f_N(t)|^2 dt \longrightarrow \frac{1}{2} (\varphi(1/4) + \varphi(3/4)).$$

Thus $|f_N|^2 dt \rightarrow m_*$ weak-star, proving $m_* \in S(2\mathbb{Z})$.

It remains to check extremality. Every polynomial with frequencies in $2\mathbb{Z}$ has squared modulus invariant under the half-translation $R(t) = t + 1/2$. Consequently every measure in $S(2\mathbb{Z})$ is R -invariant. Suppose

$$m_* = \theta \mu_1 + (1 - \theta) \mu_2, \quad 0 < \theta < 1, \quad \mu_1, \mu_2 \in S(2\mathbb{Z}).$$

Positivity implies $\mu_1, \mu_2 \ll m_*$, so both are supported on $\{1/4, 3/4\}$. Their R -invariance forces equal mass at these two points. As they are probabilities, $\mu_1 = \mu_2 = m_*$. Hence $m_* \in S_e(2\mathbb{Z})$.

Finally, (2) shows that every nonzero element of $\Sigma(F, \zeta)$ is supported on $\{0, 1/2\}$, whereas m_* is supported on $\{1/4, 3/4\}$. Therefore $m_* \notin \Sigma(F, \zeta)$, completing the counterexample. $\square$

Verification Notes

- The example is not the constant sequence $F = \mathbb{Z}$: $\chi_{2\mathbb{Z}}$ is nonconstant and its orbit is a genuine two-cycle.
- The correlation computation includes the factor $1/2$ coming from the invariant probability ζ . Since c is arbitrary, it still yields the entire nonnegative ray in (2).
- Membership of m_* in the spectral envelope is proved by an explicit norm-one polynomial sequence, avoiding the normalization ambiguity in the source’s displayed dilation formula.
- Extremality is checked in the larger simplex of half-translation invariant probabilities. It therefore implies extremality inside $S(2\mathbb{Z})$.
- No computation or asymptotic estimate beyond the standard weak-star approximate-identity property of Fejér kernels is used.

Novelty Check and Limitations

A bounded search through 22 July 2026 covered the run registry, solution, attempt, and proof-gap indexes; the local source corpus; exact searches for `S_e(F)`, `Sigma(F,zeta)`, and the wording of Question 3.4; the source title and author; and the later preliminary report arXiv:1204.4904. No prior explicit answer was found. This supports, but does not establish, originality.

The result answers the literal quantifier over minimal characteristic sequences. It does not settle the stronger aperiodic version, nor the Thue–Morse case emphasized immediately before the question. Human review should focus on whether the source intended such an unstated restriction.

References

- [1] W. Lawton, *The Feichtinger Conjecture for Exponentials*, J. Nonlinear Anal. Optim. Theory Appl. **2** (2011), no. 1, 131–140; arXiv:1012.4549.